

An endpoint-augmented lower bound for a Fourier positive-part problem

Candidate partial result for arXiv:2003.06962

11 August 2026

Status. Candidate substantial partial result, likely valid, requiring human review. The exact sharp constant remains open. The theorem below improves the source's one-dimensional lower bound from approximately 0.4107675 to 0.4796156.

1 Source problem

Madrid and Ramos study autocorrelation inequalities and a related Fourier dual problem [1]. With the convention

$$\widehat{\varphi}(\xi) = \int_{\mathbb{R}} \varphi(x) e^{-2\pi i x \xi} dx,$$

let $\varphi \in C_c^\infty(\mathbb{R})$ be even and nonnegative, with $\text{supp } \varphi \subset [-1, 1]$ and $\int_{\mathbb{R}} \varphi = 1$. Writing $u_+ = \max(u, 0)$, their Theorem 4.1 proves

$$\|(\widehat{\varphi})_+\|_1 \geq \frac{1}{2(1 + \theta_0)} = 0.410767\dots, \tag{1}$$

where $\theta_0 = -\inf_{y \in \mathbb{R}} \sin(y)/y = 0.217\dots$. Immediately afterward they state that the sharp form of this lower bound is unknown in every dimension [1, p. 17].

Equivalently, if

$$m_1 = \inf \left\{ \|(\widehat{\varphi})_+\|_1 : \begin{array}{l} \varphi \in C_c^\infty(\mathbb{R}) \text{ is even and nonnegative,} \\ \text{supp } \varphi \subset [-1, 1], \quad \int_{\mathbb{R}} \varphi = 1 \end{array} \right\},$$

then the source proves $m_1 \geq 0.410767\dots$ and asks for the sharp form. The result below proves $m_1 \geq 0.479615\dots$; it does not determine the exact infimum.

shows, however, that this attempt does not give us *any* improvement over the original result of [1].

Theorem 4.1. *Let φ be a smooth function as above. It holds that*

$$\|(\widehat{\varphi})_+\|_1 \geq \frac{1}{2(1+\theta_0)}.$$

Proof. The proof of this result resembles, in spirit, the method of Bourgain, Clozel and Kahane [3] to provide lower bounds for the root uncertainty principle in any dimension.

Indeed, we know that

$$\begin{aligned} \varphi(x) &\leq \|\widehat{\varphi}\|_1 = \|(\widehat{\varphi})_+\|_1 + \|(\widehat{\varphi})_-\|_1, \quad \forall x \in [-1, 1], \\ \varphi(0) &= \|(\widehat{\varphi})_+\|_1 - \|(\widehat{\varphi})_-\|_1. \end{aligned} \tag{4.3}$$

Also, we know that

$$\begin{aligned} 1 - 2\varphi(0) &= \int_{-1}^1 (\varphi(t) - \varphi(0)) dt = \int_{-1}^1 \left(\int_{\mathbb{R}} \widehat{\varphi}(\xi) (\cos(2\pi\xi t) - 1) d\xi \right) dt \\ &\leq \int_{-1}^1 \left(\int_{\mathbb{R}} (\widehat{\varphi})_-(\xi) (1 - \cos(2\pi\xi t)) d\xi \right) dt \\ &\leq 2(1 + \theta_0) \|(\widehat{\varphi})_-\|_1, \end{aligned} \tag{4.4}$$

so using (4.4) in the second equation in (4.3) implies that

$$\|(\widehat{\varphi})_+\|_1 \geq \frac{1}{2(1+\theta_0)} + \frac{\theta_0}{1+\theta_0} \varphi(0) \geq \frac{1}{2(1+\theta_0)},$$

as desired. $\square$

We notice that this idea does not only work in dimension one. Indeed, if one adapts the proof of Theorem 4.1 for the higher dimensional case, one obtains an asymptotic growth resembling that of Bourgain, Clozel and Kahane [3] Théorème 3] for the value of

$$\mathbb{A}_d = \inf_{f \in \mathcal{A}_d \setminus \{0\}} A(f)A(\widehat{f}),$$

where $\mathcal{A}_d$ denotes the class of even, real and integrable functions $f \in \mathbb{R}^d$ whose Fourier transform share the same properties, together with $f(0), \widehat{f}(0) \leq 0$, and

$$A(g) = \inf\{r > 0 : f(x) \geq 0, \forall |x| \geq r\}.$$

See, for instance, [7, 12, 11, 10] for sharper estimates and more recent developments in the study of the constants $\mathbb{A}_d$.

We do not know of any dimension for which we can find the sharp form of the lower bound in Theorem 4.1 but we believe that there is a strong connection between this new problem and the framework of problems we just mentioned.

Figure 1. Theorem 4.1 and the subsequent sharpness question in [1, p. 17], rendered and cropped from the local source PDF.

2 Proof intuition

The source proof uses normalization and the center value $\varphi(0)$, but it does not retain the boundary identity $\varphi(1) = 0$. Fourier inversion turns these two spatial identities into two moments of $\widehat{\varphi}$. A carefully chosen linear combination leaves the multiplier

$$K(y) = \frac{2}{3} \left(2 + \cos y - 3 \frac{\sin y}{y} \right).$$

The Cusa–Huygens inequality makes K nonnegative. Consequently only the negative part of $\widehat{\varphi}$ can increase the combined moment, exactly as needed to bound the positive mass.

The remaining issue is global: one needs a useful upper bound for K on the whole real line. Where $\sin y \geq 0$ the bound is immediate. On the first negative half-period, a tangent comparison

proves monotonicity up to $144/25$; after that point Cauchy–Schwarz controls every remaining half-period. This yields a closed-form constant. The exact sharp problem remains open because the argument uses only three scalar constraints and not the full pointwise condition $\varphi \geq 0$.

3 The multiplier estimate

Set $\text{sinc}(0) = 1$ and $\text{sinc}(y) = \sin(y)/y$ for $y \neq 0$.

Lemma 1 (Global Cusa–Huygens inequality). *For every $y \in \mathbb{R}$,*

$$\text{sinc}(y) \leq \frac{2 + \cos y}{3}. \quad (2)$$

Proof. Both sides are even. For $0 \leq y \leq \pi$, define

$$q(y) = y(2 + \cos y) - 3 \sin y.$$

Then $q(0) = 0$ and

$$q'(y) = 2 - 2 \cos y - y \sin y = 2 \sin(y/2)(2 \sin(y/2) - y \cos(y/2)) \geq 0,$$

because $\tan(y/2) \geq y/2$. Thus (2) holds on this interval. If $y \geq \pi$ and $\sin y \leq 0$, it is immediate since the right-hand side is at least $1/3$. If $\sin y > 0$, then $y \geq 2\pi$ and

$$\text{sinc}(y) \leq \frac{1}{y} \leq \frac{1}{2\pi} < \frac{1}{3} \leq \frac{2 + \cos y}{3}.$$

□

Lemma 2 (Uniform upper bound). *The nonnegative even function*

$$K(y) = \frac{2}{3}(2 + \cos y - 3 \text{sinc}(y))$$

satisfies

$$0 \leq K(y) \leq D := \frac{96 + \sqrt{2929}}{72} \quad (y \in \mathbb{R}). \quad (3)$$

Proof. Nonnegativity is Lemma 1. By evenness, take $y \geq 0$. If $\sin y \geq 0$, then $K(y) \leq 2 < D$.

Suppose $\sin y < 0$ and put

$$H(y) = \cos y + \frac{3|\sin y|}{y}.$$

On $[\pi, 3\pi/2]$ we have $H(y) \leq 3/y < 1$. We next claim that H is increasing on

$$\left[\frac{3\pi}{2}, \frac{144}{25} \right].$$

On this interval set $z = 2\pi - y$. Since both sine and cosine are positive, direct differentiation shows that $H'(y) > 0$ is equivalent to

$$\tan z > \frac{3y}{y^2 - 3}. \quad (4)$$

The difference between the two sides of (4) has derivative

$$-\sec^2(2\pi - y) + \frac{3(y^2 + 3)}{(y^2 - 3)^2} < 0,$$

because $y > 3$ implies $3(y^2 + 3) < (y^2 - 3)^2$. It is therefore enough to check the right endpoint $r = 144/25$.

The classical rational bound $\pi > 333/106$ gives

$$2\pi - r > \frac{693}{1325} =: z_0.$$

The positive Taylor coefficients of the tangent function give

$$\tan(2\pi - r) > z_0 + \frac{z_0^3}{3} + \frac{2z_0^5}{15} > \frac{3600}{6287} = \frac{3r}{r^2 - 3}.$$

For completeness, the middle strict inequality is the exact rational calculation

$$z_0 + \frac{z_0^3}{3} + \frac{2z_0^5}{15} - \frac{3600}{6287} = \frac{425878389221405094}{128378665258349609375} > 0.$$

This proves the claim.

For every $y \geq r$, Cauchy–Schwarz gives

$$H(y) \leq \sqrt{1 + \frac{9}{y^2}} \leq \sqrt{1 + \left(\frac{25}{48}\right)^2} = \frac{\sqrt{2929}}{48}.$$

The same bound holds on $[3\pi/2, r]$, because H is increasing and its endpoint value is bounded by the same Cauchy–Schwarz estimate. It also dominates the bound $H < 1$ on $[\pi, 3\pi/2]$. Hence, whenever $\sin y < 0$,

$$K(y) = \frac{2}{3}(2 + H(y)) \leq \frac{2}{3}\left(2 + \frac{\sqrt{2929}}{48}\right) = D.$$

□

4 Improved lower bound

Theorem 3. *Let $\varphi \in C_c^\infty(\mathbb{R})$ be even and nonnegative, with $\text{supp } \varphi \subset [-1, 1]$ and $\int_{\mathbb{R}} \varphi = 1$. Then*

$$\|(\widehat{\varphi})_+\|_1 \geq \frac{72}{96 + \sqrt{2929}} = 0.4796155513\dots \quad (5)$$

Proof. Put $F = \widehat{\varphi}$ and

$$P = \int_{\mathbb{R}} F_+(\xi) d\xi, \quad N = \int_{\mathbb{R}} F_-(\xi) d\xi, \quad a = \varphi(0) = P - N \geq 0.$$

The function F is real, even, and rapidly decreasing. Normalization and Fourier inversion at the endpoint give, with $y = 2\pi\xi$,

$$1 = \int_{\mathbb{R}} F(\xi) 2 \operatorname{sinc}(y) d\xi, \quad (6)$$

$$0 = \varphi(1) = \int_{\mathbb{R}} F(\xi) \cos y d\xi. \quad (7)$$

Subtracting $2/3$ times (7) from (6), and then subtracting $(4/3)a = \int F(4/3)$, yields

$$1 - \frac{4}{3}a = - \int_{\mathbb{R}} F(\xi) K(2\pi\xi) d\xi \leq DN = D(P - a),$$

where Lemma 2 was used in the last step. Since $D > 4/3$, rearrangement gives

$$DP \geq 1 + \left(D - \frac{4}{3}\right)a \geq 1.$$

Therefore $P \geq 1/D$, which is exactly (5). □

5 Verification, limitations, and search status

The proof is analytic. The accompanying `verification.md` separately checks the Fourier signs, all ranges in the multiplier estimate, and the final rearrangement. The script `verify_constant.py` reproduces the exact rational checkpoint and performs a dense numerical contradiction scan; the scan is not used to justify any global statement.

This is a partial answer to the source’s sharpness direction. It improves the known one-dimensional lower bound but does not determine the exact infimum, construct an extremizer, or treat higher dimensions. A three-parameter semi-infinite dual optimization suggests a slightly stronger certificate near 0.47995, but that computation is retained only in the attempt record because it lacks a rigorous global enclosure.

Bounded literature check. The run’s cheap indexes, local source corpus, exact title and status sentence, the source constant, and the terms “Fourier positive part”, “endpoint”, and “Cusa–Huygens” were searched on 11 August 2026. OpenAlex listed two citing works for the published article, neither addressing this sharp-bound question. The later paper [2] resolves the source’s separate average-autocorrelation compact-extremizer conjecture, not the selected Theorem 4.1 direction. No prior statement of Theorem 3 was found. This is a bounded novelty audit, not a definitive priority claim.

Human-review recommendation. First verify the sign in the identity preceding the final estimate. Then check Lemma 2’s tangent comparison, especially the monotonicity of the difference in (4). If those points pass, the packet is ready to be treated as a rigorous quantitative partial result.

References

- [1] J. Madrid and J. P. G. Ramos, *On optimal autocorrelation inequalities on the real line*, Commun. Pure Appl. Anal. **20** (2021), 369–388; arXiv:2003.06962, doi:10.3934/cpaa.2020271.
- [2] J. de Dios Pont and J. Madrid, *On classical inequalities for autocorrelations and autoconvolutions*, arXiv:2106.13873 (2021; revised 2026).